

Shengjie Zhu

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About me

I'm an Applied Scientist at Amazon Ring AI. My Ph.D. research focused on 3D vision, including depth estimation, Structure-from-Motion, image correspondence estimation, camera pose estimation, camera calibration, and etc.

Work Experience

Aug, 2024 – now **Applied Scientist II, Ring AI, Amazon.**

1. Developed [Ask Ring](#), an agentic Ring AI assistant.
 - Led Intent Routing (IR) and primary contributor to Video Query Agent (VQA).
2. Indoor & Outdoor Localization and Mapping.
 - LiDAR-Camera calibration. [\[Publication-1\]](#)
 - Indoor Localization and Reconstruction. [\[Publication-2\]](#)
3. Optimized power- and memory-efficient Ring Animal Detection models for embedded devices.

May – Aug, 2024 **Research Scientist Intern, BAIR, Google.**

Foundation depth model based Structure-from-Motion for Neural Rendering and AR/VR. [\[Publication-3\]](#)

2021 & 2022 **Applied Scientist Intern, Amazon Device AI.**

Improve the LiDAR generated depthmap quality with epipolar geometry. [\[Publication-5\]](#)

Education

2017 – 2024: **PhD, Computer Science, Michigan State University**, East Lansing, U.S.

Dissertation: Structure and Motion Estimation from Dense Depth and Correspondence

Advisor: Prof. Xiaoming Liu

2013 – 2017: **Bachelor of Engineering, Electrical and Electronics, Southeast University**, Nanjing, China.

Publications

(Under Review) **[1] Camera LiDAR Calibration in the Eye of LiDAR** [\[Website\]](#).

Shengjie Zhu, Feng Liu, Xiaoming Liu, Kah Kuen Fu, Chong Huang

(Under Review) **[2] On Self-Improving Structure-from-Motion: A Generalizable Multi-View Pose Estimator** [\[Website\]](#).

Shengjie Zhu, Ahmed Abdelkader, Yuyang Ji, Xiaoming Liu, Feng Liu, Kah Kuen Fu, Chong Huang

3DV'26 **[3] Marginalized Bundle Adjustment: Multi-View Camera Pose from MonoDepth** [\[PDF, Website\]](#).

Shengjie Zhu, Ahmed Abdelkader, Mark J. Matthews, Xiaoming Liu, and Wen-Sheng Chu

ECCV'24 **[4] Revisit Self-Supervision with Local Structure-from-Motion** [\[PDF, Website\]](#).

Shengjie Zhu, Xiaoming Liu

ECCV'24 **[5] Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry** [\[PDF, Website\]](#).

Shengjie Zhu*, Girish Chandar Ganesan*, Abhinav Kurmur, Xiaoming Liu

NeurIPS'23 **[6] Tame a Wild Camera: In-the-Wild Monocular Camera Calibration** [\[PDF, Website\]](#).

Shengjie Zhu, Abhinav Kurmur, Masa Hu, Xiaoming Liu

CVPR'23 **[7] LightedDepth: Video Depth Estimation in light of Limited Inference View Angles** [\[PDF, Website\]](#).

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CVPR'23 **[8] PMatch: Paired Masked Image Modeling for Dense Geometric Matching** [\[PDF, Website\]](#).

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CVPR'20 **[9] The Edge of Depth: Explicit Constraints between Segmentation and Depth** [\[PDF, Website\]](#).

Shengjie Zhu, Garrick Brazil, Xiaoming Liu

Computer Skills

Language **Python, C++, CUDA, Matlab, PyTorch, Tensorflow, CuPy, Numba**

Talk

Aug. 08, 2023 **3D Perception from Two Views**, Google Pixel Biometrics Seminar.

Feb. 05, 2024 **Structure-from-Motion Meets Self-supervised Learning**, CMU VASC Seminar. [\[Link\]](#)

Jul. 09, 2024 **Structure-from-Motion from Dense Learning**, Google Pixel Biometrics Seminar.